

1. Administrative & Study Identification

Full Study Title: A Study to Assess Efficacy and Safety of Pembrolizumab With or Without Sacituzumab Tirumotecan (MK- 2870) in Adult Participants With Resectable Non Small Cell Lung Cancer (NSCLC) Not Achieving Pathological Complete Response (pCR) (MK-2870-019) **Short Title/Acronym:** TroFuse-019 **Protocol Number:** MK-2870-019 **NCT Number:** NCT06312137 **Posting ID:** 265110105053023897 **Trial Status:** RECRUITING **Sponsor/Company Name:** Merck **Investigational Product:** Pembrolizumab (Keytruda) and Sacituzumab Tirumotecan (MK-2870), utilized in regimens that may include cisplatin (Platinol) **Phase of Development:** Phase III **Medical Monitor:** Clinical Director, Merck

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To assess if adding sacituzumab tirumotecan (MK-2870) with pembrolizumab after surgery is effective in treating NSCLC for participants not achieving pathological complete response. The primary hypothesis is that sacituzumab tirumotecan plus pembrolizumab is superior to pembrolizumab monotherapy with respect to disease-free survival (DFS) as assessed by blinded independent central review (BICR). * **Secondary Objectives:** To evaluate Overall Survival (OS), Distant metastasis-free survival (DMFS) as assessed by the investigator, and Disease-Free Survival (DFS) as assessed by the investigator.

2.2 Background & Rationale To evaluate whether the addition of the investigational antibody-drug conjugate sacituzumab tirumotecan (MK-2870) to standard adjuvant pembrolizumab improves survival outcomes for patients with resectable NSCLC who are at high risk of recurrence because they did not achieve a pathological complete response (pCR) following initial neoadjuvant pembrolizumab and platinum-based doublet chemotherapy (such as cisplatin).

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Pembrolizumab alone) * **Blinding:** Open-label * **Randomization:** Randomized (1:1)

3.2 Planned Enrollment & Duration * **Target \$N\$:** Approximately 780 participants randomized to the adjuvant trial treatment phase. * **Trial Status Description:** Currently recruiting participants. *

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has histological or cytological confirmation of squamous or nonsquamous non-small cell lung cancer (NSCLC), resectable clinical Stage II, IIIA or IIIB (with nodal involvement [N2]) per AJCC eighth edition guidelines. 2. Has confirmation that either epidermal growth factor receptor (EGFR)-directed or anaplastic lymphoma kinase (ALK)-directed therapy is not indicated as primary therapy. 3. Is able to undergo surgery based on the opinion of the investigator after consultation with a surgeon. 4. Is able to receive neoadjuvant pembrolizumab and platinum-based doublet chemotherapy. 5. Has not achieved pathological complete response (pCR) at surgery by local review of pathology (applicable to screening for the adjuvant period only, before randomization). 6. Tumor tissue sample from surgical resection has been provided for determination of programmed cell death ligand 1 (PD-L1) and trophoblast cell surface antigen 2 (TROP2) status by central vendor before randomization. 7. Confirmed to be disease-free based on re-baseline radiological assessment as documented by contrast-enhanced chest/abdomen/pelvis computed tomography (CT) (or MRI) within 28 days before randomization. 8. Participants with AEs due to previous anticancer therapies must have recovered to $\leq$ Grade 1 or baseline (endocrine-related AEs adequately treated with hormone replacement are eligible). 9. Controlled viral infections: HIV-infected participants must have well-controlled HIV on antiretroviral therapy (ART); HBsAg-positive participants must have received HBV antiviral therapy for $\geq$ 4 weeks with undetectable viral load; HCV-positive participants must have undetectable viral load for $\geq$ 4 weeks.

4.2 Exclusion Criteria 1. Has one of the following tumor locations/types: NSCLC involving the superior sulcus; Large cell neuro-endocrine cancer (LCNEC); Sarcomatoid tumor; Diagnosis of SCLC or, for mixed tumors, presence of small cell elements. 2. Has Grade $\geq$ 2 peripheral neuropathy. 3. Has history of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing. 4. Has active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease. 5. Has uncontrolled, significant cardiovascular or cerebrovascular disease within the past 6 months (including NYHA Class III/IV heart failure, unstable angina, MI, uncontrolled symptomatic arrhythmia, or QTcF interval $>$ 480 ms). 6. Has received prior neoadjuvant therapy for their current NSCLC diagnosis. 7. Has received prior systemic anticancer therapy (including investigational agents) within 4 weeks, or prior radiotherapy within 2 weeks of starting study intervention. 8. Has received a live or live-attenuated vaccine

within 30 days before the first dose (killed vaccines are permitted). 9. Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy (> 10 mg daily of prednisone equivalent) or any other immunosuppressive therapy within 7 days prior. 10. Has a known additional malignancy that is progressing or required active treatment within the past 5 years. 11. Has an active autoimmune disease requiring systemic treatment in the past 2 years. 12. Has a history of noninfectious pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/ILD. 13. Has an active infection requiring systemic therapy. 14. Is an HIV-infected participant with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 15. Has concurrent active Hepatitis B and Hepatitis C infection. 16. Has a history of allogeneic tissue/solid organ transplant. 17. Has not adequately recovered from major surgery or has ongoing surgical complications. 18. Severe hypersensitivity ($\geq$ Grade 3) to study intervention, its excipients, or another biologic therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: MK-2870 administered via intravenous infusion 1 time every 2 weeks, and pembrolizumab administered 1 time every 6 weeks. *** Associated Therapies:** Initial neoadjuvant phase consists of pembrolizumab plus platinum-based doublet chemotherapy, which typically includes agents such as cisplatin (Trade Name: Platinol; ATC Code: L01XA01). *** Route of Administration:** Intravenous (IV) *** Duration of Treatment:** Neoadjuvant phase for up to 12 weeks; total trial treatment and follow-up extending up to approximately 118 months.

5.2 Concomitant & Prohibited Therapy * Permitted: Standard post-operative supportive care, antiretroviral therapy (ART) for HIV, HBV antivirals, and killed vaccines. *** Prohibited:** Immunosuppressive therapy, chronic systemic steroid therapy (> 10 mg daily of prednisone equivalent), live virus vaccines within 30 days, and prior treatment with other investigational agents within 4 weeks.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Assessments, Histology Review, PD-L1/TROP2 testing
Baseline	Day 0	Surgery confirmation, pCR evaluation, Randomization, First Adjuvant Dose
Interim	Every 2 to 6 weeks	IV Infusions, Safety Labs, Efficacy Assessments
End of Study	Up to ~118 months	Final Vital Signs, Exit Interview, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Disease-free survival (DFS) as assessed by Blinded Independent Central Review (BICR), defined as the time from randomization to any recurrence (local, locoregional, regional or distant), occurrence of new primary NSCLC, or death due to any cause, whichever occurs first.

7.2 Secondary Endpoints * Overall Survival (OS). * Distant metastasis-free survival (DMFS) as assessed by the investigator.
* Disease-Free Survival (DFS) as assessed by the investigator.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection to measure the number of participants who discontinue study intervention due to AEs up to ~118 months.
 - **Serious Adverse Events (SAEs):** Reported within standard Phase 3 clinical trial timelines (typically 24 hours).
 - **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee appointed by the sponsor.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for adverse event evaluations.
 - **Sample Size Justification:** 780 participants are targeted for randomization in the adjuvant phase, sufficiently powering the study to detect a significant difference in the primary DFS endpoint.
 - **Handling of Missing Data:** Standard imputation methodologies according to the sponsor's statistical guidelines, with censoring applied for BICR assessments.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Periodic onsite and remote monitoring visits for data verification.
 - **Audit Trail:** eCRF locking, tracking of surgical outcomes, and data clarification forms via standard quality audit mechanisms.
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11. Study Identifiers & Governance

- **Primary ID:** MK-2870-019
 - **Registry Number:** NCT06312137
 - **Posting ID:** 265110105053023897
 - **Sponsor/Lead Organization:** Merck
 - **Study Title:** TroFuse-019
 - **Official Regulatory Title:** A Phase 3 Randomized Open-Label Study of Adjuvant Pembrolizumab With or Without MK-2870 in Participants With Resectable Stage II to IIIB (N2) NSCLC Not Achieving pCR After Receiving Neoadjuvant Pembrolizumab With Platinum-based Doublet Chemotherapy Followed by Surgery
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12. Clinical Framework (NSCLC Specific Adaptation)

- **Primary Condition:** Non-Small Cell Lung Cancer (NSCLC) / Non-Small Cell Lung Carcinoma
 - **Specific Sub-types Mentioned:** Metastatic Non Small Cell Lung Cancer (Metastatic non-small cell lung cancer)
 - **Diagnosis Threshold:** Resectable clinical Stage II to IIIB (N2) lacking pCR after neoadjuvant therapy
 - **Medical Methodology:** Neoadjuvant chemo-immunotherapy followed by surgery and Adjuvant Immunotherapy + Antibody-Drug Conjugate
 - **Optimization Target:** Disease-Free Survival (DFS) prolongation
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13. Study Procedures & Methodology

- **Management Pathway:** Standard surgical oncology clinical pathway combined with adjuvant infusions
 - **Education Protocol (HCP):** Site visits and investigator meetings on MK-2870 administration and AE management.
 - **Education Protocol (Patient):** Informed consent, treatment scheduling, and AE self-reporting guidance.
 - **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of imaging and tumor pathology.
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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to ~118 months
 - **Onsite Visit Cadence:** Infusions scheduled every 2 weeks (for MK-2870) and every 6 weeks (for pembrolizumab).
 - **Remote Monitoring Frequency:** Regular interval contacts for survival tracking.
 - **Checklist Review Interval:** Routine scheduled intervals for tumor imaging/response assessments per RECIST guidelines.
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15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					